# Ali Haider

Charlotte, NC

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## Summary

Senior Data Scientist with 5+ years of experience delivering end-to-end machine learning solutions across finance, retail, and cybersecurity. Adept at building scalable ML pipelines, predictive models, fraud detections, and decision-support tools in production environments. Skilled in Python, PyTorch, TensorFlow, and SQL, with deep expertise in time-series forecasting and anomaly detection. Experienced in A/B testing, data exploration, and working with large-scale distributed systems. Proven ability to collaborate with product, engineering, and business stakeholders to turn complex data problems into actionable insights that drive operational efficiency and customer satisfaction.

## Experience

**Founding AI/ML Engineer** — Remote

Apply Sprint - (Acquired) — *Sep 2025 – March 2026*

- Built and shipped a production-grade multi-agent AI system (LangGraph + LangChain) for intelligent job matching and resume optimization, powered by structured LLM orchestration (GPT-4, Claude, DeepSeek). Designed a two-stage NLP pipeline, fine-tuned XLM-RoBERTa (PyTorch) as an upstream classifier, feeding a downstream stacked ensemble with classical NLP feature engineering, deployed end-to-end on AWS via Dockerized containers.
- Led and managed a small team of engineers across architecture, ML pipeline design, and production deployment as Founding Engineer on an acquired product.

**Senior Data Scientist** — Charlotte, North Carolina

Bank of America — *Feb 2024 - Aug 2025*

- Led end-to-end model validation and ongoing monitoring for large-scale classification models supporting Mastercard and OMNI commercial credit card portfolios, covering $20B+ in annual transactions. Monitored model performance using CSI, PSI, AUC-ROC, and rank ordering, ensuring stability and regulatory compliance.
- Conducted deep statistical root cause analysis across dozens of KPIs, and developed surrogate fraud detection models using black-box distillation techniques to gain insights into the third party models.
- Authored executive-facing model documentation and risk reports communicating performance findings and remediation strategies to senior leadership and cross-functional stakeholders.

**Data Scientist** — Charlotte, North Carolina

Lowe's — *Feb 2022 – Jan 2024*

- Media Mix Modeling: Designed and implemented a robust statistical framework to evaluate marketing effectiveness. Built a baseline sales model using XGBoost to isolate incremental impact, tackled multicollinearity, and applied ensemble and Bayesian (PyMC3) methods for accurate attribution. Conducted extensive data analysis and experiments to validate assumptions, handle lag effects, and overcome data sparsity. Resulted in a 3% uplift in ROI on a $900M annual marketing spend.
- Sales Prediction During Extreme Weather: Trained, tested, and deployed regression models to forecast sales across 3,000 stores and 30,000 assortments during severe weather disruptions. Used K-Means clustering to group similar stores and product categories, enabling tailored modeling strategies per cluster. Integrated weather data, historical sales, and store-level metadata to enhance feature engineering. Leveraged scikit-learn, XGBoost, and hyperparameter tuning for model optimization. Deployed full end-to-end ML pipeline using Docker, Kubernetes, and MLflow for experiment tracking and model registry, with CI/CD managed via GitHub. Improved inventory planning, reduced stockouts, and boosted customer satisfaction through data-driven decisions.

**Data Scientist** — Irvine, California

HTN Networks Inc Cisco — *Apr 2020 – Jan 2022*

- Developed and implemented a reinforcement learning and ensemble machine learning model to identify malicious network traffic. Continuously improved model performance through experimentation and testing. Established robust post-deployment checks, including data drift monitoring, to guarantee ongoing effectiveness.
- Improved model accuracy through rigorous testing, leading to precise threat identification. Implemented post - deployment checks ensuring model reliability and ongoing effectiveness, achieving a 97% F1 score, reducing false positives by 42%, and improving response time by 35%.

**Graduate Research Assistant** Raleigh, North Carolina
NC State University *Aug 2017 – Jan 2018*

- Contributed to a sustainability research lab, applied ANOVA and ARIMA for trend forecasting, enhancing prediction reliability to analyze environmental data trends for environmental impact assessments.
- Utilized statistical methods to derive insights, improving environmental impact assessments.

**Teaching Assistant** Istanbul, Turkey
Istanbul Sehir University *Sep 2015 – May 2017*

- Supported undergraduate instruction through grading, tutoring, and leading discussion sessions.
- Led Python and SQL lab sessions, improving student coding proficiency and understanding for assignments related to industrial engineering and Machine learning.

## Education

**North Carolina State University** Raleigh, NC
Master of Science in Operation Research *Dec 2020*

**Istanbul Sehir University** Istanbul, Turkey
Master of Science in Industrial & System Engineering *Dec 2017*

**NED University of Eng and Tech.** Karachi, Pakistan
Bachelor of Engineering in Industrial engineering *Dec 2013*

## Key Skills

Machine Learning, Predictive Modeling, XGBoost, PyTorch, Python (Pandas, NumPy, SciPy, Matplotlib), Advanced SQL, Snowflake, A/B Testing, Time-Series Forecasting, Statistical Analysis, Feature Engineering, Model Evaluation, Spatial Analysis, Geospatial Data, Data Fusion, Airflow, AWS, CI/CD, Git, Simulation, ETA Prediction, Communication (Technical Writing and Presentations), Bayesian Modeling (PyMC3, PyStan)

## Self-Initiated Projects

- Developed a multivariate time-series forecasting model for stock prices using LSTM and sentiment signals extracted via NLP from financial news headlines. Implemented using TensorFlow and NLTK.
- Trained a convolutional neural network (CNN) on custom image datasets to classify grocery items with real-world camera images. Used data augmentation and batch normalization to improve generalization.
- Built a Transformer-based chatbot capable of context retention and semantic understanding using Hugging Face and spaCy; deployed via a lightweight Flask web interface.
- Used a conditional Generative Adversarial Network (GAN) to augment imbalanced datasets for classification tasks, improving model accuracy on minority classes by 15%.
- Implemented a Deep Q-Network (DQN) agent for simulating equity trading strategies using OpenAI Gym and PyTorch, integrating reward shaping and experience replay for improved learning efficiency.

## Publications

Authored 7 peer-reviewed publications in reputable journals, leveraging statistical methods and data science techniques to extract insights and solve real-world problems. Full list available at: **Google Scholar:** https://scholar.google.com/citations?user=tq8Rh54AAAAJ&hl=en

## Honors & Awards

2017 **Recogonized with NC State PhD fellowship award**, A prestigious award granted to a select few pursuing a PhD, recognizing exceptional academic excellence and research potential. *NC, USA*